# Multidomain Preisach Hysteresis Results for Metal / Interlayer / AlScN / Metal Stacks

Krishna Chemudupati

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**Abstract**

This report documents the active multidomain Preisach hysteresis workflow in the present repository and summarizes the main trends that emerge from the generated figure set. The model computes quasistatic polarization–voltage loops for metal / interlayer / AlScN / metal stacks by combining a multidomain Preisach representation of the ferroelectric, one-dimensional electrostatics with finite metal screening, and a self-consistent bisection solver. The main results are organized around three questions: how the loop changes with interlayer thickness, how the choice of interlayer material affects loop retention, and how sensitive the loop is to electrode boundary conditions. The discussion is kept close to the active figures and tables so that each modeling conclusion is connected directly to a generated result.

## Contents

# 1 Purpose of the model

The active workflow is designed to answer a specific materials question: how the ferroelectric hysteresis loop changes when the interlayer thickness, interlayer material, electrode pair, and AlScN thickness are varied. Throughout this report the plotted quantity is the net ferroelectric polarization of the AlScN layer. It is therefore the average polarization of the multidomain state after the stack electrostatics and the domain switching have been solved self-consistently. It is not the screening charge, the electric displacement, or a measured current.

At a single applied voltage $V_a$, the model guesses an internal field acting on the ferroelectric, determines which domains would switch under that field, computes the corresponding net polarization $P_{\mathrm{FE}}$, passes that polarization into the electrostatic stack model, and then recomputes the field that the stack itself would produce. The guessed field is adjusted until the field used to switch the domains agrees with the field returned by the electrostatics. The hysteresis loop is the collection of those self-consistent solutions along the prescribed voltage sweep.

# 2 Baseline active workflow

The active interlayer-thickness figures use a fixed multidomain setup with 15000 domains, a structural disorder distribution centered at $c/a = 1.54$ with standard deviation $\sigma_{c/a} = 0.04$, the constitutive maps

$$P_s(c/a) = 333.33(c/a) - 400 \quad \mu\mathrm{C/cm}^2$$

and

$$E_c(c/a) = 3.16(c/a) - 1.1 \quad \mathrm{MV/cm},$$

random initial domain states generated from a fixed seed, a quasistatic bisection solver, and a polarization-space stopping tolerance of 0.5 $\mu$C/cm$^2$. Each run consists of two voltage cycles, although only the final cycle is plotted. The interlayer sweep used throughout the active figure set is

$$0,\ 0.1,\ 0.2,\ 0.5,\ 0.8,\ 1,\ 2,\ 3\ \mathrm{nm}.$$

The dominant electrostatic ingredients are the top and bottom electrode screening lengths, the top and bottom work functions, the interlayer thickness and dielectric constant, the ferroelectric thickness and dielectric constant, and the instantaneous ferroelectric polarization. In the active code the screening charge is computed from

$$\sigma_s = \frac{\dfrac{P_d t_d}{k_d} + \dfrac{P_{\mathrm{FE}} t_f}{k_f} + \epsilon_0 V_a}{\dfrac{\lambda_t}{k_t} + \dfrac{\lambda_b}{k_b} + \dfrac{t_d}{k_d} + \dfrac{t_f}{k_f} + \dfrac{t_i}{k_i}}, \tag{1}$$

and the internal ferroelectric field is then written as

$$E_{\mathrm{FE}} = \frac{\sigma_s - P_{\mathrm{FE}}}{k_f \epsilon_0} + \frac{\Delta V_{\mathrm{bi,FE}}}{t_f}. \tag{2}$$

These two relations are enough to explain the trends seen later in the figures. A thicker interlayer weakens the loop because it both reduces how effectively the applied voltage reaches the ferroelectric and increases the electrostatic penalty associated with polarization. A higher-$k$ interlayer weakens the loop less strongly because it is a better dielectric match to the ferroelectric. Poorer metal screening reduces the stable polarization on both sides of the loop, while work-function asymmetry introduces a horizontal preference through the built-in term.

## 3 $Al_2O_3$ interlayer sweeps

To keep the sweeps informative rather than trivially saturated, the active figure set uses a different voltage window for each AlScN thickness. The 45 nm case is run at ±20 V, the 20 nm case at ±12 V, and the 10 nm case at ±6.5 V. The resulting $Al_2O_3$ interlayer sweeps are shown in Figures 1, 2, and 3.

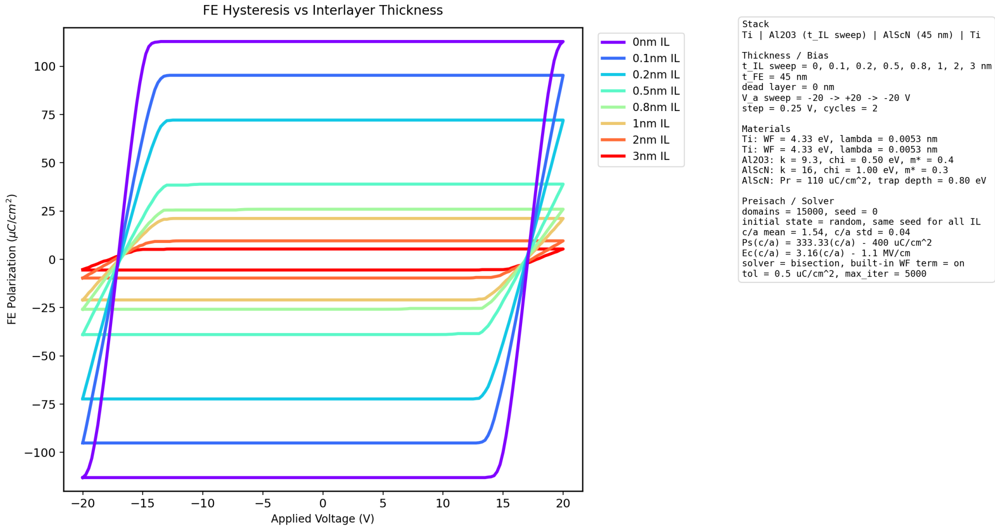


Figure 1: $Ti/Al_2O_3/AlScN/Ti$ IL-thickness sweep for 45 nm AlScN at ±20 V.

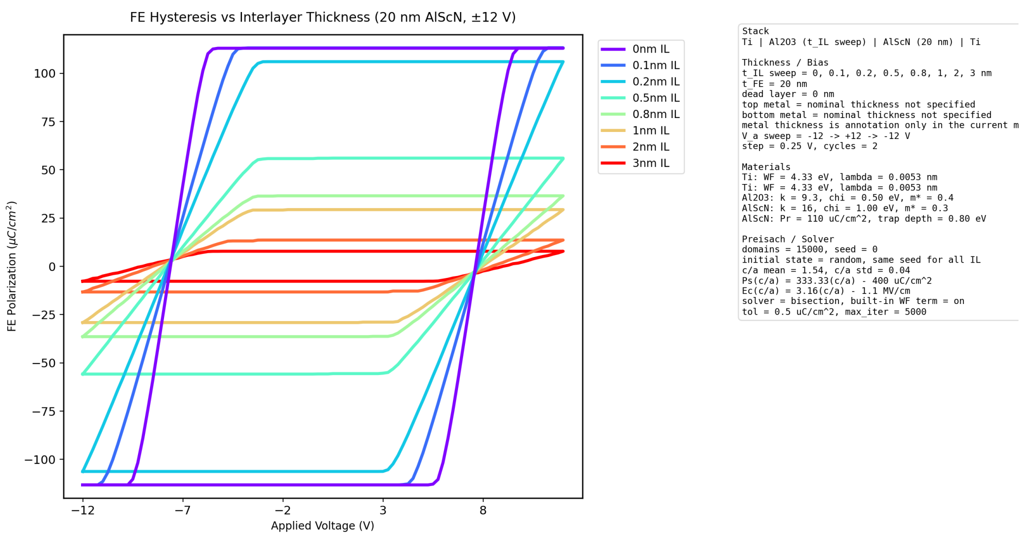


Figure 2: $Ti/Al_2O_3/AlScN/Ti$ IL-thickness sweep for 20 nm AlScN at ±12 V.

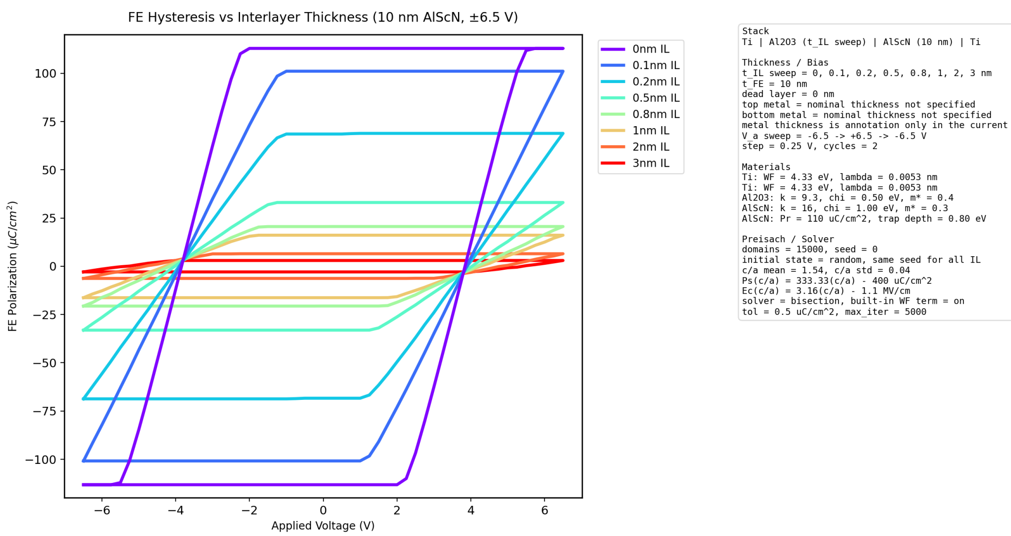


Figure 3: $Ti/Al_2O_3/AlScN/Ti$ IL-thickness sweep for 10 nm AlScN at ±6.5 V.

Table 1 summarizes the positive-branch amplitudes extracted from these loops.

Table 1: Positive-branch $P_{\max}$ values for Ti/$Al_2O_3$/AlScN/Ti loops ($\mu$C/cm$^2$).

| Case | 0 nm | 0.1 nm | 0.2 nm | 0.5 nm | 0.8 nm | 1 nm | 2 nm | 3 nm |
|---|---|---|---|---|---|---|---|---|
| 45 nm, ±20 V | 112.835 | 95.340 | 72.146 | 38.986 | 25.992 | 21.203 | 9.600 | 5.380 |
| 20 nm, ±12 V | 112.977 | 113.167 | 106.027 | 56.070 | 36.548 | 29.397 | 13.630 | 7.830 |
| 10 nm, ±6.5 V | 112.895 | 101.079 | 68.863 | 33.053 | 20.600 | 16.104 | 6.498 | 3.003 |

Several points are worth noting. At 0 nm IL all three thickness cases are essentially fully polarized, with amplitudes very close to ±113 $\mu$C/cm$^2$. Once the IL is introduced, the loops shrink rapidly in all three cases. The 45 nm case shows the strongest suppression at large IL for the chosen voltage window and falls to 5.380 $\mu$C/cm$^2$ at 3 nm IL. The 20 nm case remains the most robust of the three under its chosen drive and still reaches 7.830 $\mu$C/cm$^2$ at 3 nm IL. The 10 nm case at ±6.5 V is more clearly voltage-limited and falls to 3.003 $\mu$C/cm$^2$ at 3 nm IL. The important point is that a thinner ferroelectric is not simply "more sensitive" in a single universal sense. A thinner film is easier to drive at fixed voltage, but it is also more vulnerable to depolarization. The apparent thickness trend therefore depends on the voltage window as well as the geometry itself.

# 4 Replacing $Al_2O_3$ with $HfO_x$

The same thickness-specific voltage windows were then applied to Ti/$HfO_x$/AlScN/Ti. The resulting sweeps are shown in Figures 4, 5, and 6.

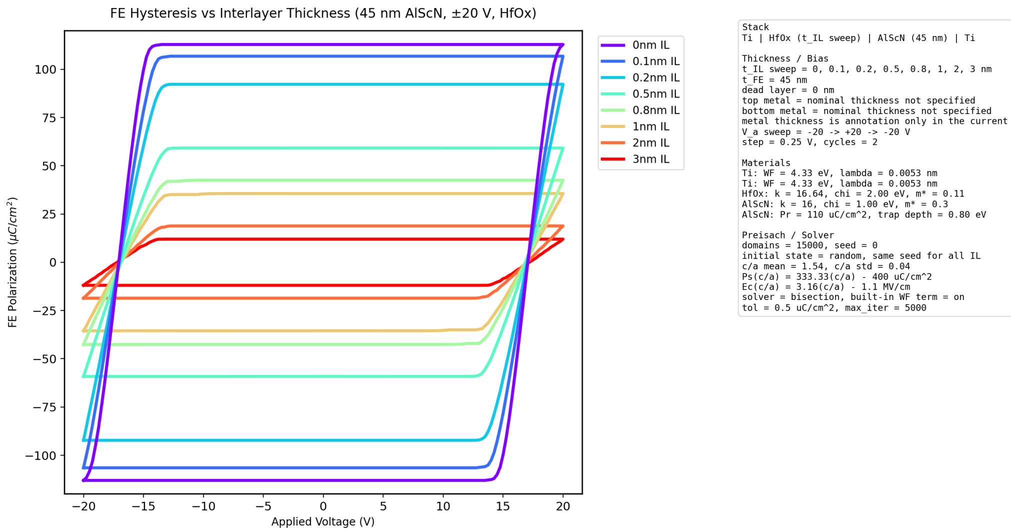


Figure 4: Ti/HfO$_x$/AlScN/Ti IL-thickness sweep for 45 nm AlScN at ±20 V.

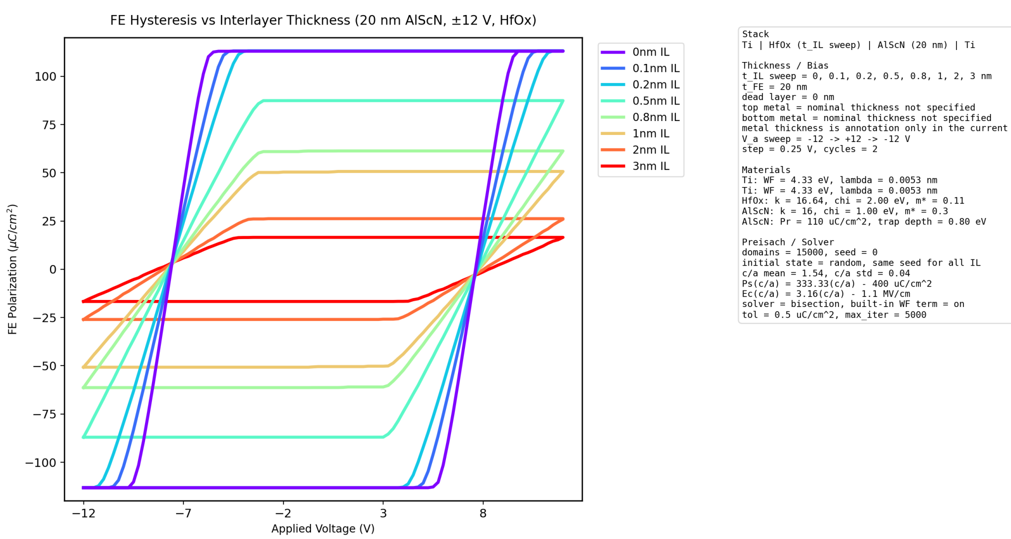


Figure 5: Ti/HfO$_x$/AlScN/Ti IL-thickness sweep for 20 nm AlScN at ±12 V.

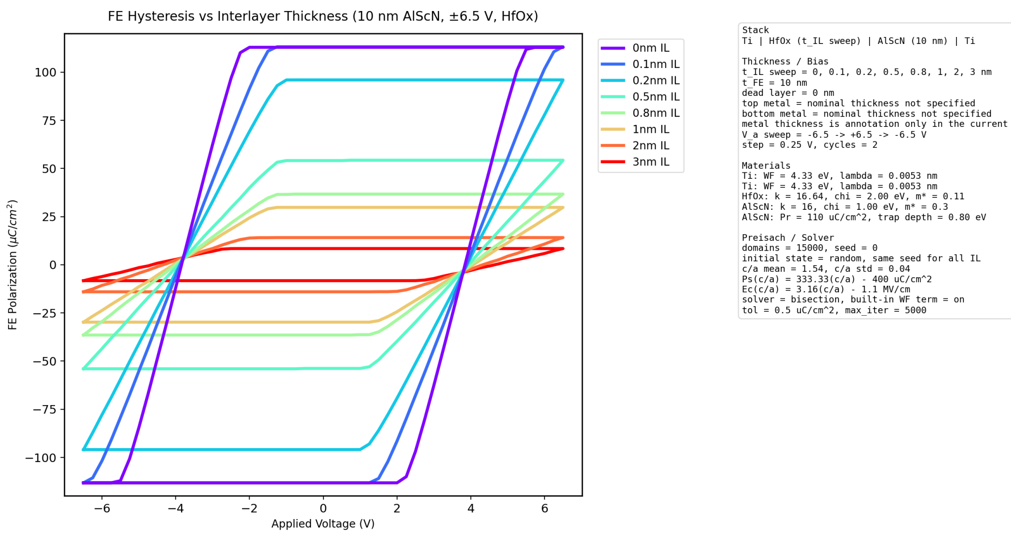


Figure 6: Ti/$HfO_x$/AlScN/Ti IL-thickness sweep for 10 nm AlScN at ±6.5 V.

Table 2: Positive-branch $P_{\max}$ values for Ti/$HfO_x$/AlScN/Ti loops ($\mu$C/cm$^2$).

| Case | 0 nm | 0.1 nm | 0.2 nm | 0.5 nm | 0.8 nm | 1 nm | 2 nm | 3 nm |
|---|---|---|---|---|---|---|---|---|
| 45 nm, ±20 V | 112.835 | 106.786 | 92.224 | 59.221 | 42.542 | 35.610 | 18.843 | 11.999 |
| 20 nm, ±12 V | 112.977 | 113.167 | 113.167 | 87.398 | 61.313 | 50.687 | 26.230 | 16.533 |
| 10 nm, ±6.5 V | 112.895 | 113.188 | 95.970 | 54.278 | 36.693 | 29.793 | 14.119 | 8.426 |

The dominant result is that, for the same geometry and voltage window, $HfO_x$ preserves much larger loops than $Al_2O_3$ at finite IL thickness. This follows directly from the active dielectric constants: $k_{\mathrm{AlScN}} = 16$, $k_{\mathrm{Al_2O_3}} = 9.3$, and $k_{\mathrm{HfO_x}} = 16.64$. $HfO_x$ is therefore a much better electrostatic match to AlScN than $Al_2O_3$. In the present model this means that less voltage is lost across the IL and the depolarization penalty is weaker. The effect is substantial. For the 45 nm case at 3 nm IL, the $HfO_x$ loop reaches 11.999 $\mu$C/cm$^2$ while the $Al_2O_3$ loop reaches only 5.380 $\mu$C/cm$^2$. For the 20 nm case at the same IL thickness, $HfO_x$ gives 16.533 $\mu$C/cm$^2$ while $Al_2O_3$ gives 7.830 $\mu$C/cm$^2$. For the 10 nm case the corresponding values are 8.426 $\mu$C/cm$^2$ and 3.003 $\mu$C/cm$^2$. In other words, switching from $Al_2O_3$ to $HfO_x$ roughly doubles the large-IL loop amplitude in the thicker two cases and improves it by almost a factor of three in the 10 nm, lower-voltage case.

## 5 Electrode sensitivity at 0 nm IL

The repository also includes a 0 nm IL electrode study, which is useful because it isolates the electrode boundary condition from interlayer-material effects. The resulting overlay and summary heatmap are shown in Figures 7 and 8.

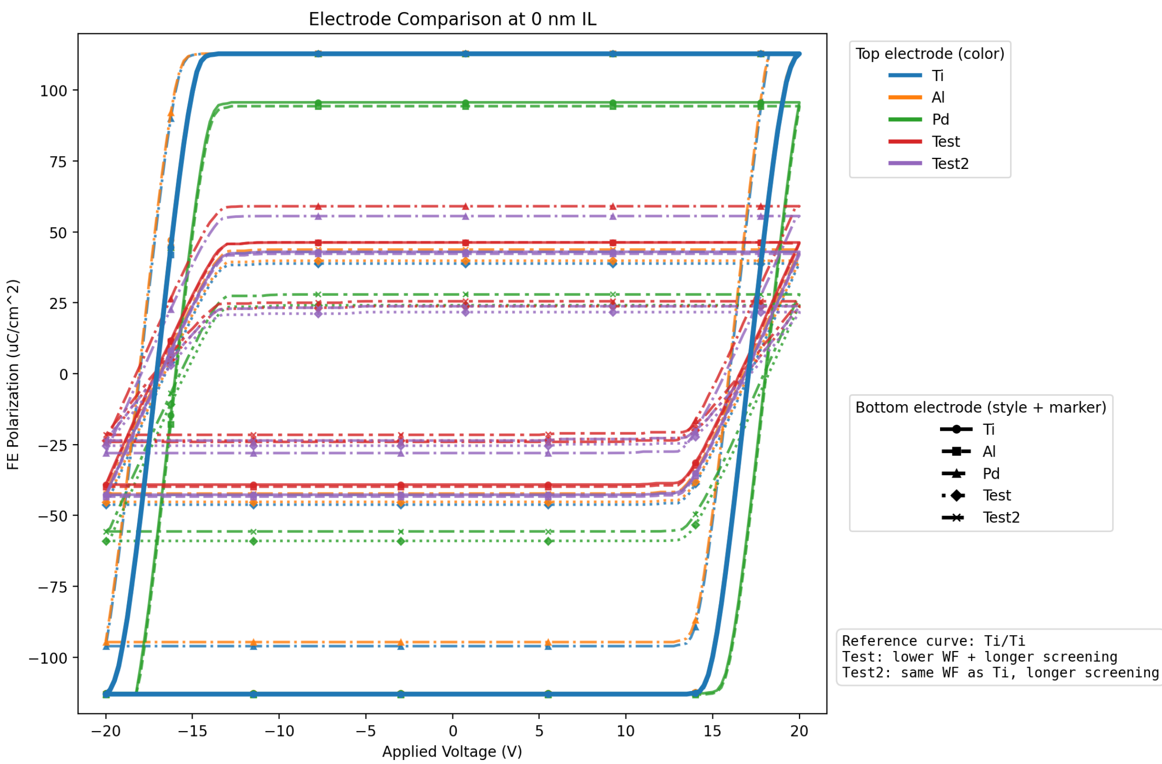


Figure 7: Overlay of all ordered top/bottom electrode combinations at 0 nm IL. Color denotes the top electrode, while line style and marker denote the bottom electrode.

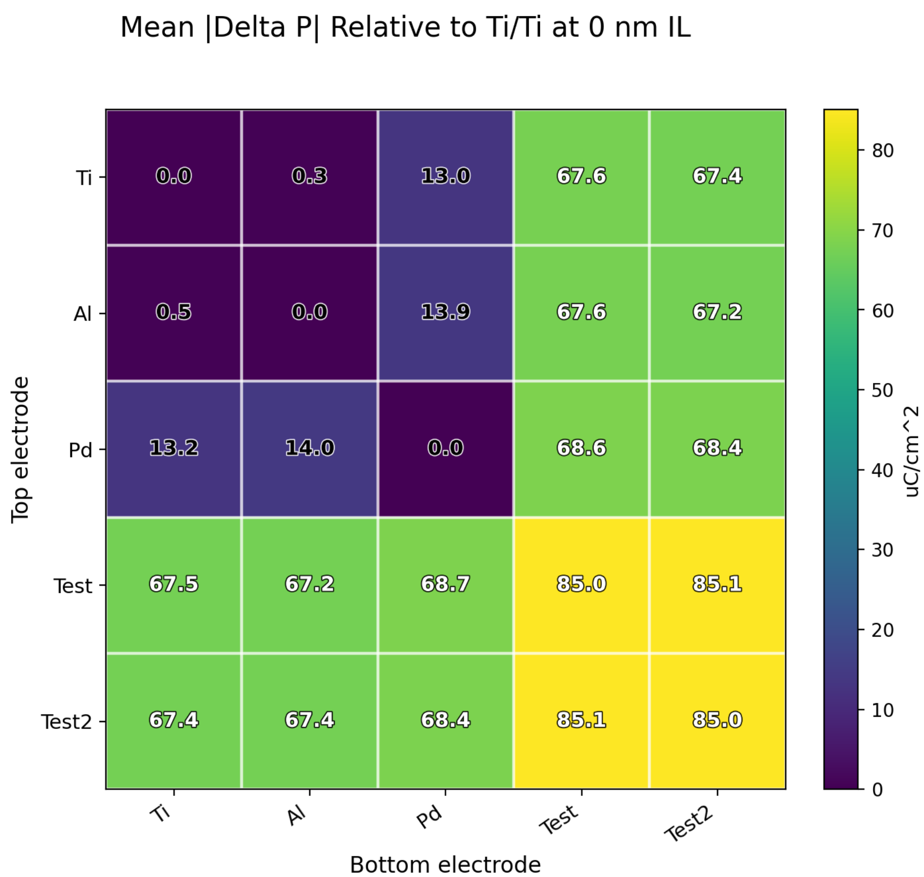


Figure 8: Mean absolute polarization difference relative to Ti/Ti at 0 nm IL.

Table 3: Representative 0 nm IL electrode effects relative to Ti/Ti.

| Stack | mean $\vert\Delta P\vert$ ($\mu$C/cm$^2$) | max $\vert\Delta P\vert$ ($\mu$C/cm$^2$) | $\Delta P(0\text{V}, \text{fwd})$ | $\Delta P(0\text{V}, \text{rev})$ |
|---|---|---|---|---|
| Ti/Al | 0.343 | 2.933 | -0.102 | 0.041 |
| Ti/Pd | 13.014 | 59.715 | 16.930 | -0.041 |
| Pd/Ti | 13.193 | 59.642 | 0.298 | -17.135 |
| Ti/Test | 67.623 | 84.359 | 66.823 | -73.976 |
| Ti/Test2 | 67.365 | 80.180 | 70.329 | -69.851 |

This comparison separates two physically distinct effects. The first is work-function asymmetry. Ti/Al remains close to Ti/Ti because the work-function mismatch is small in the active parameter set. Ti/Pd and Pd/Ti differ much more strongly, and those two stacks reverse one another in the way expected from a built-in field that changes sign when the electrode order is reversed. The second effect is screening-length sensitivity. The most revealing comparison is Ti/Test2. Test2 was designed to keep the Ti work function while increasing the screening length, and the fact that Ti/Test2 still differs strongly from Ti/Ti shows that, in the present hysteresis-only implementation, electrode screening length can change the loop just as strongly as work-function asymmetry. This is useful because it shows that the metal boundary condition is not determined by work function alone; it also depends on how effectively the electrodes compensate ferroelectric bound charge.

## 6 Interpretation of the generated loop shapes

Some of the generated loops are more boxy or rhomboid than the smooth textbook shape often associated with ferroelectrics. In the present model this is a natural consequence of the assumptions rather than a plotting artifact. Each domain is binary rather than continuously varying, switching is threshold-based and abrupt, the solver is quasistatic rather than dynamical, and the internal field can vary almost linearly with applied voltage over part of the sweep. When a large collection of domains is driven through a fairly ordered set of thresholds under those conditions, the net polarization can follow straight segments and produce the boxier loop shape seen in some figures.

## 7 Scope and limitations

The current repository already accounts coherently for several important trends. It captures the strong suppression caused by increasing interlayer thickness, it shows why a higher-$k$ interlayer preserves the loop more effectively, it resolves loop skew introduced by work-function asymmetry, and it demonstrates that poorer electrode screening can shrink the loop even when the work function is held fixed. Within the present workflow these trends are emergent outputs of the coupled equations once the geometry, material set, and sweep protocol have been fixed.

At the same time, this remains a compact hysteresis model rather than a full ferroelectric-diode transport simulator. It does not yet include DC current transport, thermionic emission, tunneling, trap-assisted current mechanisms, explicit Landau–Khalatnikov dynamics, time-dependent switching kinetics, spatial domain-wall physics, or finite metal thickness in the electrostatics.

## 8 Concluding remarks

The active multidomain Preisach workflow supports a clear set of conclusions. The hysteresis curves are not drawn by hand or imposed by post-processing; they are produced by a coupled memory-plus-electrostatics model. Interlayer thickness is a dominant control variable because it enters the electrostatic denominator directly. $HfO_x$ is considerably more favorable than $Al_2O_3$ for preserving loop amplitude at finite IL thickness in the present model. Electrode screening length is a major hysteresis lever alongside work-function asymmetry. Finally, thickness trends cannot be interpreted independently of the chosen voltage window, because the ease of driving the ferroelectric and the depolarization penalty compete throughout the sweep.